CS204 – Algorithms & Data Structures Laboratory

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Tutorial #05 C++ Inheritance

Overview

- Introduction
- Variety of Inheritances
- Inheritance Types And Access Rules
- Examples
- Inheritance & Constructors
- Base Class Member Functions & Their Behaviors
- Type Casting
- Functions & Base Class Objects
- Inheritance & Operator Overloading

Introduction

- Sometimes, we want to express commonality between classes
- We want to create new class from existing class by adding new members or replacing (i.e., hiding/overriding) existing members
- Inheritance is a form of software reuse
- We create a class that absorbs an existing class's data and behaviors and enhances them with new capabilities.
- The new class should inherit the members of an existing class
- This existing class is called the base class
- The new class is referred to as the derived class.

Introduction

- A derived class represents a more specialized group of objects
- A derived class contains behaviors inherited from its base class and can contain additional behaviors.
- A derived class can also customize behaviors inherited from the base class.
- A direct base class the base class from which a derived class explicitly inherits
- An indirect base class inherited from two or more levels up in the class hierarchy.
- In the case of single inheritance a class is derived from one base class.

Introduction

- In the case of multiple inheritance, a derived class inherits from multiple (possibly unrelated) base classes
- public, protected and private inheritance are possible
- Variety of inheritance.

Variety of Inheritances

Variety of inheritances

Single Inheritance

```
class A
{
  private:
    int a1, a2, a3;
  protected:
    float p;
  public:
    A();
    ~A();
    f();
};
class B : public A
{
  private:
    int b1, b2, b3;
  public:
    B();
};
```

Variety of inheritances

Single Inheritance

```
class B
{
  private:
    int a1, a2, a3;
    Int b1, b2, b3;
  protected:
    float p;
  public:
    f();
    B();
};
```

A's members are all inherited
A's member functions are inherited

Variety of inheritances

Single Inheritance

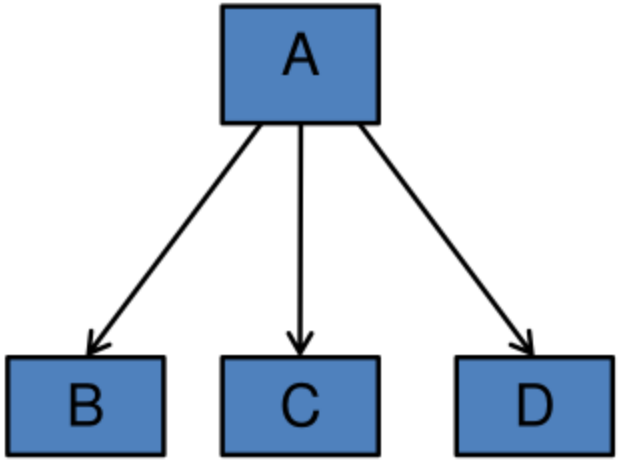
```
class A
{

};

class B : protected A
{

};
```

Variety of inheritances

Single Inheritance

```
class A
{

};

class B : private A
{

};
```

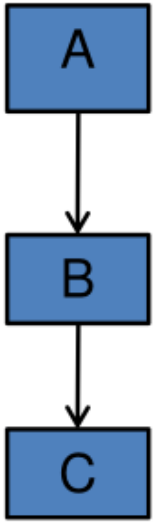
Variety of inheritances



Hierarchical Inheritance

```
class A
{

};

class B : public A
{

};
class C : public A
{

};

class D : public A
{

};
```

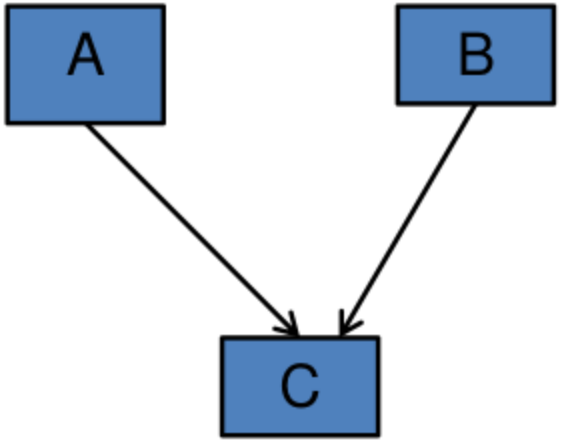
Variety of inheritances



Multilevel Inheritance

```
class A
{

};

class B : public A
{

};
class C : public B
{

};
```

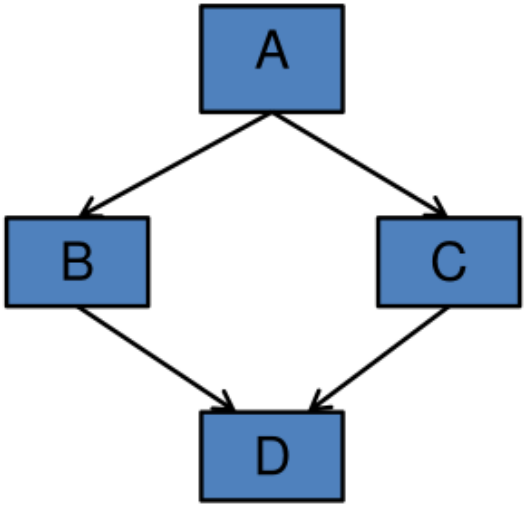
Variety of inheritances



Multiple Inheritance

```
class A
{

};

class B
{

};
class C : public B, public A
{

};
```

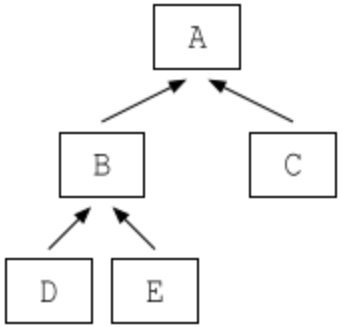
Variety of inheritances



Hybrid Inheritance

```
class A
{

};

class B : public A
{

};
class C : public A
{

};
class D : public B, public C
{

};
```

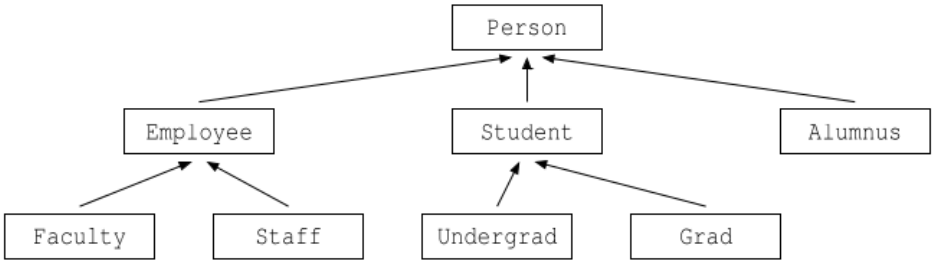
Variety of inheritances




```
class A
{

};
```

```
class B : public A
{

};
class C : public A
{

};
class D : public B
{

};
class E : public B
{

};
```

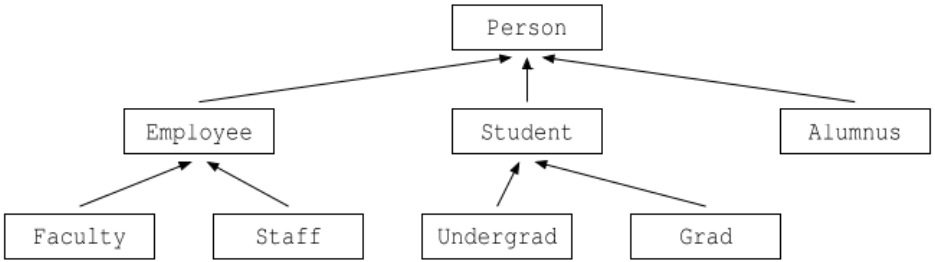
Variety of inheritances




```
class Person
{

};
```

```
class Employee : public Person
{

};
class Student : public Person
{

};
class Alumnus : public Person
{

};
```

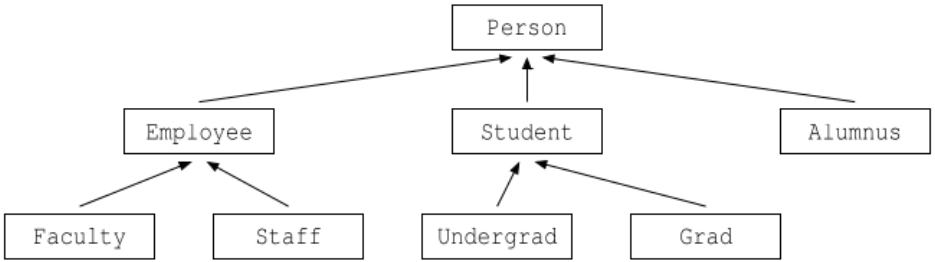
Variety of inheritances




```
class Faculty: public Employee
{

};
class Staff : public Employee
{

};
```

```
class Employee : public Person
{

};
```

Variety of inheritances




```
class Undergrad : public Student
{

};
class Grad : public Student
{

};
```

```
class Student : public Person
{

};
```

Inheritance Types And Access Rules

public inheritance

```
class Base
{
  public:
     void f();
  protected:
     void g();
  private:
     int x;
};

class Derived : public Base
{
  // f() is public
  // g() is protected
  // x is private and inaccessible from Derived
}
```

protected inheritance

```
class Base
{
  public:
     void f();
  protected:
     void g();
  private:
     int x;
};

class Derived : protected Base
{
  // f() is protected
  // g() is protected
  // x is private and inaccessible from Derived
}
```

private inheritance

```
class Base
{
  public:
     void f();
  protected:
     void g();
  private:
     int x;
};

class Derived : private Base
{
  // f() is private
  // g() is private
  // x is private and inaccessible from Derived
}
```

Examples

public inheritance

```
class Widget : public Derived
{
  public:
     void func_4()
     {
        func_2(); // OK
     }
};

int main(argc, char *argv[])
{
  Derived d;
  d.func_1(); // OK
  d.func_2(); // Error: protected
  d.x_0 = 10; // Error in accessible
}
```

```
class Base
{
  public:
     void func_1();
  protected:
     void func_2();
  private:
     int x_;
};

class Derived : public Base
{
  Public:
   Void func_3()
   {
     func_1(); // OK
     func_2(); // OK
     x_ = 0; // Error x_0: is private to Base
   }
}
```

protected inheritance

```
class Widget : public Derived
{
  public:
     void func_4()
     {
        func_2(); // OK
     }
};

int main(argc, char *argv[])
{
  Derived d;
  d.func_1(); // Error: protected
  d.func_2(); // Error: protected
  d.x_0 = 10; // Error: private
}
```

```
class Base
{
  public:
     void func_1();
  protected:
     void func_2();
  private:
     int x_;
};

class Derived : protected Base
{
  Public:
   Void func_3()
   {
     func_1(); // OK
     func_2(); // OK
     x_ = 0; // Error: x_0 is private to Base
   }
}
```

private inheritance

```
class Widget : public Derived
{
  public:
     void func_4()
     {
        func_2(); // Error: inaccessible
     }
};

int main(argc, char *argv[])
{
  Derived d; // OK. Public constructor
  d.func_1(); // Error: inaccessible
  d.func_2(); // Error: inaccessible
  d.x_0 = 10; // Error: inaccessible
}
```

```
class Base
{
  public:
     void func_1();
  protected:
     void func_2();
  private:
     int x_;
};

class Derived : private Base
{
  Public:
   Void func_3()
   {
     func_1(); // OK
     func_2(); // OK
     x_ = 0; // Error: x_0 is private to Base
   }
}
```

Inheritance & Constructors

Inheritance & constructors

- Constructors are not inherited

```
class A
{
  private:
    int a1, a2, a3;
  protected:
    float p;
  public:
    A();
    ~A();
    f();
};
class B : public A
{
  private:
    int b1, b2, b3;
  public:
    B();
};
```

```
class B
{
  private:
    int a1, a2, a3;
    Int b1, b2, b3;
  protected:
    float p;
  public:
    f();
    B();
};
```

Inheritance & constructors

- New members of derived class are not included in the base class
- That is b1, b2, b3 new data members in the derived class B are not part of class A.
- The data members a1, a2, a3 and p are inherited in class B
- Similarly, member function f() is inherited in class B

```
class A
{
  private:
    int a1, a2, a3;
  protected:
    float p;
  public:
    A();
    ~A();
    f();
};
class B : public A
{
  private:
    int b1, b2, b3;
  public:
    B();
};
```

Inheritance & constructors

- Since base-class constructors cannot initialize derived-class data members, inheriting constructors from base class by default would be bad idea
- This could lead to uninitialized data members
- For example, data members a1, a2, a3 and p are initialized in default constructor A().
- Class B has no constructor. Data members b1, b2 and b3 could go uninitialized and base class members are all initialized.
- A() cannot initialize b1, b2, b3

```
class A
{
  private:
    int a1, a2, a3;
  protected:
    float p;
  public:
    A(){a1=a2=a3=0; p=0.0}
    f();
};
class B : public A
{
  private:
    int b1, b2, b3;
  public:
};
```

Inheritance & constructors

- Default constructor, copy constructor and move constructor cannot be inherited
- Copy constructor A cannot be inherited

```
class A
{
  private:
    int a1, a2, a3;
  protected:
    float p;
  public:
    A(const A&)
    {
       a1=A.a1; a2=A.a2; a3=A.a3;
    }
    f();
};
class B : public A
{
  private:
    int b1, b2, b3;
  public:
};
```

Inheritance & constructors

- To inherit all other constructors, using statement must be invoked.

```
class B : public A
{
  public:
    int b1, b2, b3;
  // Following constructors NOT inherited
  // A(int k, int j)
  // A(int i, int j, int k)
};
int main(int argc, char *argv[])
{
  // Result in compilation error
  B box_01(10, 20, 30);
  B box_02(10, 20);
}
```

```
class A
{
  private:
    int a1, a2, a3;
  public:
    A(int k, int j)
    {
       a1 = k; a2 = j;
       a3 = k;
    }
    A(int i, int j, int k)
    {
      a1=i; a2=j; a3=k;
    }
};
```

Explanation

- Base class A has two constructors A(int, int) and A(int, int, int)
- Derived class B does not have constructor
- By default, class A constructors are NOT inherited
- Therefore, class B has no constructor
- In main the statement B box_01(10, 20, 30); tries to declare object box_01 with a constructor which takes three arguments
- class B having no constructors this statement is illegal and result in compilation error

Explanation

- Base class A has two constructors A(int, int) and A(int, int, int)
- Derived class B does not have constructor
- By default, class A constructors are NOT inherited
- Therefore, class B has no constructor
- In main the statement B box_01(10, 20); tries to declare object box_01 with a constructor which takes two arguments
- class B having no constructors this statement is illegal and result in compilation error

Inheritance & constructors

- To inherit all other constructors, using statement must be invoked.

```
class B : public A
{
  public:
    int b1, b2, b3;
  using A::A;
  // Following constructors are INHERITED
  // A(int k, int j)
  // A(int i, int j, int k)
};
int main(int argc, char *argv[])
{
  // OK
  B box_01(10, 20, 30);
  B box_02(10, 20);
}
```

```
class A
{
  private:
    int a1, a2, a3;
  public:
    A(int k, int j)
    {
       a1 = k; a2 = j;
       a3 = k;
    }
    A(int i, int j, int k)
    {
      a1=i; a2=j; a3=k;
    }
};
```

Explanation

- Base class A has two constructors A(int, int) and A(int, int, int)
- Derived class B does not have constructor
- "using A::A;" statement explicitly inherit A(int, int) and A(int, int, int) in class B
- Therefore, class B has two constructors from **base class**
- In main the statement B box_01(10, 20, 30); tries to declare object box_01 with a constructor which takes three arguments.
- class B has no constructor with three arguments; however inherited constructor A(int, int, int) is available.
- Therefore box_01 object is successful in creation. It calls base constructor A(int, int, int). Initializes a1, a2 and a3.
- The data members b1, b2 and b3 goes uninitialized

Constructors in both classes

- Writing identical constructors in both base class and derived class
- That is A(int, int, int) and B(int, int, int) as well as
- A(int, int) and B(int, int) is as follows
- This will initialize private members a1, a2 and a3 as well as b1, b2 and b3.

Constructors in both classes

```
class B : public A
{
  private:
    int b1, b2, b3;
  public:
    B(int k, int j) : A(k, j)
    {
      b1 = k; b2 = j;
      b3 = k;
    }
    B(int i, int j, int k):A(i,j,k)
    {
      b1 = i; b2 = j; b3 = k;
    }
};
```

```
class A
{
  private:
    int a1, a2, a3;
  public:
    A(int k, int j)
    {
       a1 = k; a2 = j;
       a3 = k;
    }
    A(int i, int j, int k)
    {
      a1=i; a2=j; a3=k;
    }
};
```

Constructors in both classes

```
int main(int argc, char *argv[])
{
  B box_01(10, 20, 30);
}
```

This leads to calling constructor of class B with three arguments B(int, int, int)

B(int, int, int) in turn calls A(int, int, int) explicitly and initializes private data members of class A

Followed by private data members of class B are initialized

Base Class Member Functions & Their Behavior

Base class member functions and their behavior

```
class Base
{
  // Assume members
  public:
   void print()
   {
     cout << "a1: " << a1 << " ";
     cout << "a2: " << a2 << " ";
     cout << "a3: " << a3 << endl;
   }
};
```

```
class Derived : public Base
{
  // Assume members
  public:
  // print() is not defined here
};

int main(int argc, char *argv[])
{
  Derived d;
  d.print();
}
```

As no print() member function is defined in the Derived class, Base class print() function is inherited

In the main function, a call to print through d.print() invokes base class print function

Base class member functions and their behavior

```
class Base
{
  // Assume members
  public:
   void print()
   {
     cout << "a1: " << a1 << " ";
     cout << "a2: " << a2 << " ";
     cout << "a3: " << a3 << endl;
   }
};
```

```
class Derived : public Base
{
  public:
  void print()
  {
     cout << "b1: " << b1 << " ";
     cout << "b2: " << b2 << " ";
     cout << "b3: " << b3 << endl;
  }

};

int main(int argc, char *argv[])
{
  Derived d;
  d.print();
}
```

Base class member functions and their behavior

- The print() function is implemented in both Base and Derived classes
- Depending on the invoked object, appropriate function is called
- Observed the following example

```
int main(int argc, char *argv[])
{
  Derived d;
  d.print(); //Derived print();

  Base b;
  b.print(); //Base print();
}
```

Base class member functions and their behavior

```
class Fruit
{
  public:
   void print()
   {
     cout << "Fruit class" << endl;
   }
};

class Apple : public Fruit
{
  public:
   void print()
   {
     cout << "Apple class" << endl;
   }
};
```

```
class Banana : public Fruit
{
  public:
   void print()
   {
     cout << "Banana class" << endl;
   }
};
```

```
int main(int argc, char *argv[])
{
  Fruit f; f.print(); //Fruit::print
  Apple a; a.print(); //Apple::print
  Banana b; b.print(); //Banana::print
}
```

Type Casting

Type Casting

- **The statement** `int j = 12; float f = (float) j;`
- **Variable j is of type int**
- **It is type casted to float**
- **That is integer 12 to floating point 12**
- **Resulting value is assigned to variable f**
- **You will not observe any issues in general but for extreme values.**

Base Class Object Assigned To Derived Class Object

Type Casting

- The following type casting is not allowed in classes
- `Example Base b; Derived d = b;`
- This is not allowed. It results in compilation error.
- The reason is that the `Base` class object `b` does not have the members d1, d2, d3 of `Derived` class.
- Therefore, there's no meaningful way to copy `b` into `d` because the derived class-specific members cannot be initialized from `b`.
- The compiler sees that `b` is of type `Base`, and `d` is of type `Derived`. Since there's no implicit way to convert a `Base` object into a `Derived` object, this operation is not allowed.

Type Casting

```
#include<iostream>
using namespace std;
class Base
{
  private:
    int b1, b2, b3;
  public:
    Base()
    {
      b1 = 10; b2 = 20; b3 = 30;
    }
    void print()
    {
      cout << "b1: " << b1 << " ";
      cout << "b2: " << b2 << " ";
      cout << "b3: " << b3 << endl;
    }
};
```

```
class Derived: public Base
{
  private:
    int d1, d2, d3;
  public:
    Derived()
    {
      d1 = 100; d2 = 200; d3 = 300;
    }
    void print()
    {
      cout << "d1: " << d1 << " ";
      cout << "d2: " << d2 << " ";
      cout << "d3: " << d3 << endl;
    }
};
```

Type Casting

```
int main(int argc, char *argv[])
{
  Base b;
  b.print();
  Derived d;
  d = b;
  d.print();
}
```

This results in compilation errors.

Type Casting

```
int main(int argc, char *argv[])
{
  Base b; b.print();
  Derived* dp = static_cast<Derived*>(&b);
  dp->print();
}
```

```
saradhi@Betaal:~/courses/cs204/july-dec-2024/tutorials/week06$ g++ inheritance-05.cc -g
saradhi@Betaal:~/courses/cs204/july-dec-2024/tutorials/week06$ ./a.out
b1: 10 b2: 20 b3: 30
d1: -741619200 d2: -926382803 d3: 2060400720
saradhi@Betaal:~/courses/cs204/july-dec-2024/tutorials/week06$
```

You can force the type casting but results in un-initialized values

Type Casting – Correct Method

```
class Derived: public Base
{
  private:
    int d1, d2, d3;
  public:
    Derived()
    {
      d1 = 100; d2 = 200; d3 = 300;
    }
    Derived(const Base &b)
    {
      d1 = 1000; d2 = 2000; d3 = 3000;
    }
    void print()
    {
      cout << "d1: " << d1 << " ";
      cout << "d2: " << d2 << " ";
      cout << "d3: " << d3 << endl;
    }
};
```

A type conversion constructor must be defined in the Derived class

The construct accepts a Base class reference.

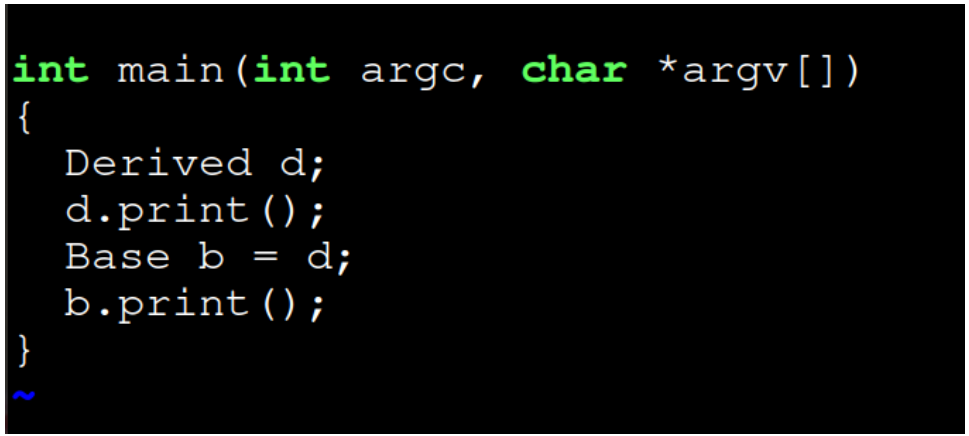
The assignment
Derived d = b;

works as this conversion constructor is called during the assignment

```
int main(int argc, char *argv[])
{
  Base b; b.print();
  Derived d = b;
  d.print();
}
```

Derived Class Object Assigned To Base Class Object

Type Casting

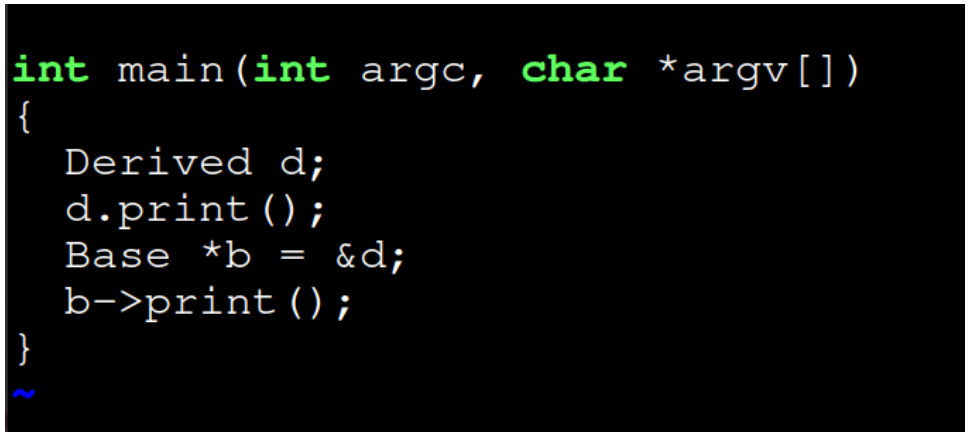


```
int main(int argc, char *argv[])
{
  Derived d;
  d.print();
  Base b = d;
  b.print();
}
~
```

The above type casting will work.

Assigning derived class object to base class object will work

Type Casting



```
int main(int argc, char *argv[])
{
  Derived d;
  d.print();
  Base *b = &d;
  b->print();
}
~
```

The above type casting will also work.

Assigning derived class object to base class object will work

Functions And Base Class Objects

Functions & Type Casting

```
void function_1(Base &b)
{
  b.print();
}

int main(int argc, char *argv[])
{
  Base b;
  Derived d;

  function_1(b);
  function_1(d);
}
```

The above type casting will also work.

Assigning derived class object to base class object will work

Functions & Type Casting

```
void function_2(Base *b)
{
  b->print();
}

int main(int argc, char *argv[])
{
  Base *b = new Base();
  Derived *d = new Derived();

  function_2(b);
  function_2(d);
}
```

The above type casting will also work.

Assigning derived class object to base class object will work

Inheritance & Operator Overloading

Inheritance & Operator Overloading

- Overloaded operators are inherited by derived classes in C++, just like any other member function.
- If you define an overloaded operator as a member function in a base class, it will be inherited by any derived class.
- This means that the derived class can use that operator as if it were defined in the derived class itself.
- The inherited operator will behave according to the logic defined in the base class.
- However, the operator will work with the data members of the derived class if it is applied to an instance of the derived class.

Inheritance & Operator Overloading

- Just like any other member function, you can override an overloaded operator in a derived class by defining the operator function in the derived class.

- If the derived class overrides an operator but still wants to use the base class's version, it can explicitly call the base class's operator using the scope resolution operator (Base::operator=).

Inheritance & Operator Overloading

```cpp
#include <iostream>
using namespace std;

class Base
{
  public:
    int x;

    Base(int val) : x(val)
    {
    }
    Base operator+(const Base& other) const
    {
        return Base(x + other.x);
    }

    void display() const
    {
        cout << "Base x: " << x << endl;
    }
};
```

```cpp
class Derived : public Base
{
  public:
    int y;

    Derived(int v1, int v2) : Base(v1), y(v
    {

    }

    Derived operator+(const Derived& d) con
    {
        return Derived(x + d.x, y + d.y);
    }

    void display() const
    {
        cout << "Derived x: " << x << ", y:
    }
};
```

Inheritance & Operator Overloading

```
int main()
{
    Base b1(5), b2(10);
    Base b3 = b1 + b2;
    b3.display();

    Derived d1(3, 7);
    Derived d2(4, 6);
    Derived d3 = d1 + d2;
    d3.display();

    Base b4 = d1 + d2;
    b4.display();

    return 0;
}
```

Base b3 = b1 + b2;

Calls Base class operator+ function

Derived d3 = d1 + d2;
Calls Derived class operator+ function

Base b4 = d1 + d2
Calls Derived class operator and assigns only x value in b4

Questions?

Thank You